

CS698B: Fundamentals of Data Engineering (Part I)

2026-27-I, IIT Kanpur – Problem Sheet 1



General Instructions:

1. This problem sheet need not be submitted.
2. The questions are meant for practice and self-assessment.
3. Please use this resource honestly to assess your understanding of concepts.

Q1 (Become the Machine). Write the output of the following code snippet exactly, down to whitespaces.

```
x = 7 // 2
y = 7 / 2
print( x, type( x ), end = ':' )
print( y, type( y ), end = ':' )
```

Q2 (Become the Machine). Write the output of the following code snippet exactly, down to whitespaces.

```
s = "12" + "3"
n = int( "12" ) + int( "3" )
print( s )
print( n )
```

Q3 (Complete the Code). Complete the comprehension so `squares` becomes `[ 0, 1, 4, 9, 16 ]`

```
squares = [ _____ for i in range( 5 ) ]
```

Q4 (Become the Machine). Write the output of the following code snippet exactly, down to whitespaces.

```
a = [ 1, 2, 3 ]
b = a
b.append( 4 )
print( a )
print( b )
```

Q5 (Become the Judge). The following code is meant to print `True` if `n` is prime (for `n > 0`) else print `False`. The code has a bug. Design a valid test case to reveal the bug where the code gives wrong output.

```
n = int( input() )
is_prime = True
for i in range( 2, n ):
    if n % i == 0:
        is_prime = False
print( is_prime )
```

Q6 (Become the Machine). Write the output of the following code snippet exactly, down to whitespaces.

```
grid = [ [ 1, 2 ], [ 3, 4 ] ]
print( [ x for row in grid for x in row ] )
```